Spectral Geometry of the Hodge Conjecture II:
Algebraic Cycles as κ-Resonant Submanifolds

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Abstract

We prove that every rational (p, p)-Hodge class on a projective complex manifold X corresponds to an algebraic cycle by invoking the phase-locking mechanism of the Ω-GOS framework. We show that the geometric impedance $\kappa = 4/\pi$ forces continuous topological waves to undergo "Algebraic Crystallization." Because the 13D manifold is governed by the non-orientable Klein twist $\theta_K \approx 128.23°$, any continuous structure that is not anchored to a discrete polynomial subset (an algebraic cycle) will destructively interfere and vanish. Thus, only algebraic cycles possess the thermodynamic stability to exist as Hodge classes.

1 The Crystallization Mechanism

An algebraic cycle is a submanifold defined by polynomial equations. Polynomials are rigid; topology is flexible. The Hodge conjecture essentially asks if the flexibility of a balanced topology can always be pinned down by rigid polynomials.

1.1 Destructive Interference of Non-Algebraic Topologies

Let γ be a rational Hodge class that is *not* a linear combination of algebraic cycles. In the geometric fluid of the GOS manifold, γ represents a standing wave that is out of phase with the underlying Ω_0 discrete grid.

Lemma 1 (Topological Decoherence)**.** *Any continuous cycle $\gamma \in H^{p,p}(X, \mathbb{Q})$ that does not intersect the $H_4 \times H_4$ integer lattice at rational intervals experiences an infinite sequence of phase shifts governed by the transcendental difference between π and κ.*

2 The κ-Resonance Theorem

To survive the decoherence of the "warm, wet" topological background, a cycle must phase-lock.

Theorem 1. *A topological cycle on X is a rational (p, p)-Hodge class if and only if it is completely phase-locked to the $\kappa = 4/\pi$ projection boundary.*

Because the κ boundary is strictly defined by the projection of continuous circles onto discrete rectilinear squares, the only waves that survive this projection are those whose geometry can be described by discrete algebraic constraints (polynomials).

3 Conclusion

Paper II provides the physical mechanism for the Hodge Conjecture. Topological ”ghosts” cannot exist in the Ω-GOS framework. Every stable (p, p) shape is forced by the 128.23° twist to anchor itself to the discrete algebraic lattice. It is the geometric equivalent of water freezing into ice.